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Seaport- dry port network design considering multimodal transport and carbon emissions

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ABSTRACT

Dry port can speed the flow of cargo between ships and major land transportation networks while reducing carbon emissions and relieving seaport congestion. This study introduces a continuous approximation model to facilitate the design of a seaport-dry-port network that takes carbon emissions into account. Multimodal transport with two transport modes: rail and road are considered. Containers are delivered quickly from shippers to dry ports through road transport; then transferred to seaport through rail transport to save money and reduce carbon emission. For decision-making scheme, a game theory is used to help the seaport, dry ports, and shippers optimize their own benefits and guide them to determine their decision variables. A nonlinear optimization technique is used to solve this seaport-dry-port network design problem. The closed-form of all decision variables is found directly. A numerical study is conducted to derive managerial insights. The results show that the development of dry port concepts and multimodal transports could reduce the carbon cost from road transport.

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1. Introduction

The increasing importance of environmental consciousness intensifies the challenges for today's business. Developing the transport systems that take into account the growth of trade, environmental considerations, community restrictions as well as the evolution of freight transportation and logistics is an essential first step in effective environmental management. In transportation systems, the rail freight is usually cheaper than the road freight as train can carry larger volumes over longer distances. In addition, railroad transports are four times more fuel efficient than road transportation (AoAR, 2017). According to Environmental Protection Agency (EPA) data, freight railroads account for just 0.6 per cent of U.S. greenhouse gas emissions from all sources and just 2.5 per cent of emissions from transportation-related sources (EPA, 2016). However, rail transports are lack of flexible and convenient, as routes and times cannot be adjusted. Road transport tends to be more expensive but offers much flexibility in terms of final destination and volume of goods to be transported. Thus, multimodal transport which combining the best parts of both rail-road services to suit the customers' requirements is common in today business.

The developing of dry port concepts involves the multimodal transport's idea. Containers are delivered quickly from shippers to dry ports through road transport; then transferred to seaport through railroad to save money and reduce carbon emission. Leveque and Roso (2002) defined a dry port as "an inland intermodal terminal directly connected to seaport(s) with high capacity transport mean(s), where customers can leave/pick up their standardized units as if directly to a seaport". A dry port may also serve as a depot for the storage of containers, as well as a management point responsible for integration, transhipment, temporary storage, consolidation, distribution, and customer's clearance. A well-designed dry port can benefit the ecological environment and the quality of life by shifting flows from road to rail (Roso, 2007). Dry port concept contributes to reducing the carbon emission from container ships industry. However, the development of dry ports also brings some problems about cost, location, profits, management. Apply the supply chain management perspective to consider the seaport-dry port network problems become more important and useful in both practice and research.

This paper introduces a continuous approximation (CA) model to the design of a seaport-dry-port network under the consideration of multimodal transport and carbon emissions. The research makes the following contributions to the literature. This is the first study to use the CA approach in the design of a seaport-dry-port

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network. This study answers the following questions: how many dry ports should be opened? Which shipper should be served by which dry port? The proposed model is able to solve big data size problems through finding closed or near-closed form solutions directly, which is crucial for the management of installations in the real world. Secondly, this study also uses a game theoretical decision-making scheme for a seaport, dry ports, and shippers in the network. Each player determines its decision variables to optimize its own benefit while shares the cost with others. Thirdly, this study considers the carbon emission cost and the two social costs, accidents (fatalities, injuries, and property damage) and noise, into our models. They are defined as the external costs of rail and road transportations. Finally, some observations and managerial insights are made from numerical analysis. For example, this study guides government/company in making their decisions when other costs (holding cost, carbon emission cost, transportation cost…) change.

In the next section, we present a literature review covering the topics of CA, dry ports, multimodal transport, and carbon emissions. The seaport-dry-port network problem is defined and modelled in Section 3. Section 4 describes the proposed solution procedure. In Section 5, managerial insights are provided based on the results of a numerical analysis. Conclusions are drawn in Section 6.

2. Literature review

2.1. Continuous approximation (CA)

CA models are based on spatial (and/or temporal) density and the average distribution of demand, rather than on precise information pertaining to every demand point. Discrete models are able to provide managers with optimal solutions in simple cases; however, when data and computational requirements increase, continuous models offer more realistic descriptions (Dasci and Verter, 2001). Continuous approximation can be used to supplement mathematical programming models, due to reduced data requirements and closed or near closed-form solutions.

The idea of applying continuum mechanic techniques to finite-dimensional operations-research-type problems was first introduced by Newell (1973). This study modified the problems to include continuous functions, based on the fact that final continuum problems are trivial enough to be solved explicitly using elementary calculus. Instead of trying to determine the actual location of every point, it instead seeks only to determine the approximate optimal density of these points. The results also showed that in some cases, the approximate non-unique solution of an idealized problem may be more useful than the (possibly unique) “exact” solution. This method is utilized in the logistics systems proposed by Langevin et al. (1996). Dasci and Verter (2001) developed Daganzo’s idea which used a continuous model for production-distribution system design. The study also showed the efficiency of continuous model compared to mathematical model. Recently, Murat et al. (2010) presented a continuous analysis framework to solve location-allocation problems with dense demand. Tsao et al. (2012) and Tsao and Lu (2012) modelled a multi-echelon, single-item supply chain using a two-phase CA approach. Campbell (2013) used a CA model for time-defined many-to-many transportation. Tsao et al. (2015) designed a multi-item distribution network problem through the application of CA. Tsao et al. (2017) used a continuous approximation to model a supply chain network with product manufacturing. They proved that the CA approach is useful and suitable to apply in many different fields of network design. This paper tries to use CA approach to address the seaport-dry port network design problems.

2.2. Dry ports

In recent years, increases in the flow of shipping containers have led to congestion and delays in seaports. The development of inland ports as a solution to seaport space problems has been much-discussed in the maritime context (Rytkonen, 1999). Early research by Slack (1990) and Slack (1999) demonstrated the value of inland ports and illustrated the role they play in reducing the environmental effects of multimodal transport. Thus, dry ports have been investigated by many researchers. Leveque and Roso (2002) defined a dry port as an interior multimodal terminal linked directly to a seaport, where clients are able to pick up and leave containers, as at a seaport. Roso et al. (2009) extended the theory behind the dry port concepts and defined three dry port categories: distance, midrange and closed.

This has made the location of dry ports an important issue in research. Xu (1999) considered a discrete choice model for locating an inland container depot, with the objective of maximizing profit. Yang (2005) applied the method of multi-criteria decision-making to locating dry ports in Texas, USA. Wang et al. (2009) applied a fuzzy c-average value clustering method in dry port selection. Wang and Wei (2008) used analytic network process (ANP) to evaluate location selection of the dry port at Tianjin, China. Ma (2009) presented a cellular automata program for locating dry ports serving the port of Tianjin, China. Li et al. (2011) proposed affinity propagation (AP) clustering to aid in the location planning of dry ports. Fang et al. (2012) analysed influencing factors in predicting the location of dry ports based on a logit model. Li et al. (2014) researched the topic from the perspective of AHP. Feng et al. (2013) constructed a location-allocation model for the regional seaport-dry-port network optimization problem and developed a greedy algorithm to obtain its solution. In their model, links connect the origins of freight to seaports, either directly or through dry ports. Crainic et al. (2015) modelled a dry-port-based freight distribution plan and applied mixed-integer programming to solve it. Nguyen and Notteboom (2016) presented a conceptual framework for the inclusion of multiple criteria in the evaluation of dry port location in developing countries from a multiple stakeholder perspective. Wang et al. (2017) developed a discrete model to obtain the location of dry ports. Their proposed model was applied to the port Tianjin in China. In the above research, there is lack of mathematical models which consider the seaport-dry port location problems. This paper proposes mathematical models to address the seaport-dry port network design problems. The findings of the study could help government or companies to develop the strategic plan for sustainable development in the future.

2.3. Multimodal transport

Coping with the environmental impacts will require the transportation industry and its customers to move from a traditional transportation system into an environmental transportation model. Multimodal transportation which combines the best part of several different modes of transport (e.g. rail, sea, road) plays an important role in reducing the carbon emission from transportation. Transport of goods in container could be moved on land by rail or truck & on water by ship or barge (TRB, 1998). Luo and Grigalunas (2003) summarized the development and application of a spatial-economic, multimodal container transportation demand simulation model for major US container ports. Limbourg and Jourquin (2009) proposed an iterative procedure based on both the *p*-hub median problem and the multi-modal assignment problem. They tried to optimize the rail-road container terminal locations on the European network. Liu and Yang (2015) addressed joint slot allocation and dynamic pricing for multi-node container sea-rail

multimodal transport from the revenue management point of make-to-order by applying a two-stage optimal model. Sun et al. (2017) proposed a TOPSIS decision model based on entropy weight to effectively select the optimal multimodal transport plan. A case study of Ningbo-Zhoushan port was considered to illustrate the model. In our model, a multimodal transport is applied in which containers are transferred to seaport by using rail and road transports. Shippers could choose the route through rail or road transport to minimize their cost.

2.4. Carbon emissions

Container ships are the fastest growing segment of marine shipping in terms of carbon emissions (Ocean Policy Research Foundation, 2008). However, only a few studies have considered container shipping incorporating carbon emission. Corbett et al. (2009) evaluated whether vessel speed reduction is a cost-effective means of mitigating CO_2 emissions. Hanaoka and Regmi (2011) concluded that by encouraging a modal shift, dry ports would help to ease road traffic congestion and reduce emissions. Lättilä et al. (2013) reported a method of estimating potential locations for dry ports, analysing the impact of dry ports in Finland, and determining to what degree dry ports can lower the emissions of transportation. Their results were similar to the findings of Roso (2007), who estimated that a dry port is capable of decreasing CO_2 emissions by 25%. Qiu et al. (2015) considered a hinterland supply chain with one dry port and multiple shippers located in close proximity, from an environmental perspective. In that study, carbon emissions were calculated for shared as well as independent transportation systems in order to maximize the profit of the dry port and minimize the total cost incurred by shippers. The savings in CO_2 emissions resulting from the sharing of transport could increase with the number of shippers within a certain range. Tran et al. (2017) designed a container shipping route considering the cost of shipping, inland/feeder transport, inventory, and carbon emission. A discrete mathematical programing was used to address their problems. In a different way, this paper introduces a continuous approximation (CA) model to the design of a seaport-dry-port network under the consideration of carbon emissions.

3. A seaport-dry port network design

3.1. Assumptions and notations

This paper designs a network including a seaport, several dry ports, and many shippers. In our model, a seaport is supposed to serve for a large service area, S. The seaport service area is given. Dry ports are responsible for integration, transhipment, temporary storage, consolidation and distribution, and customs clearance. Containers could be delivered directly to the seaport or through dry ports (Fig. 1).

The grid cover-couple approach (Tsao et al., 2012) is used to divide the network area into clusters. The grid-cover helps divide the seaport service area$S$, into regions based on shipper's demand. The regions with similar demand rate can be clustered together. We suppose $C_1, C_2, \ldots, C_N$be the clusters within S. For example, there are three clusters C_1, C_2, and C_3in Fig. 2. In each cluster, some dry ports should be opened.

Each dry port serves for a service area. The size of the service area for all dry ports in clusterC_i is similar and equal toA_i. The dry port service area has somewhat irregular shapes as opposed to circles, hexagons, or squares. This irregularity is shown to have little impact on the optimal solution (Dasci and Verter, 2001). Let f_r be a constant depending on the distance metric and shape of the dry port service region, the value of f_r varies on the different shape of service area. In our model, the dry port's service area is assumed to be close to circular and dry port is located in the center of the service area (Fig. 3). The average distance travelled by item is $f_r\sqrt{A_i}$, $f_r = 0.01$ (Dasci and Verter, 2001). Dry ports determine service areas to maximize the total network profit. Thus, the total number of dry ports in each cluster can be calculated (C_i/A_i) base on the service area for a dry port in cluster i (A_i) and the service area for each cluster (C_i).

The shippers' demand in each cluster i per unit time, is represented by a function of storage price, $D_i = a_i - b_i P_i$, where a_i and $b_i$are the coefficients of demand function ($a_i$is the maximum demand, $a_i > 0$; $b_i$is the price elasticity of demand, $b_i > 0$). The demand function is used to describe the situation, where "the price goes up, the demand goes down" described in the literature review (Whitin, 1955; De Borger et al., 2008). In our model, shippers could choose to route freight to a seaport, either directly or through a dry port. The shipper demand for direct route to seaport is $D_i^S = a_i^S - b_i^S P_i^S$ and the shipper demand for transferring containers through dry port is$D_i^D = a_i^D - b_i^D P_i^D$.

This paper further describes a seaport-dry-port network using game theory, wherein the seaport is treated as a leader, dry ports as the first follower, and shippers as the second followers and each

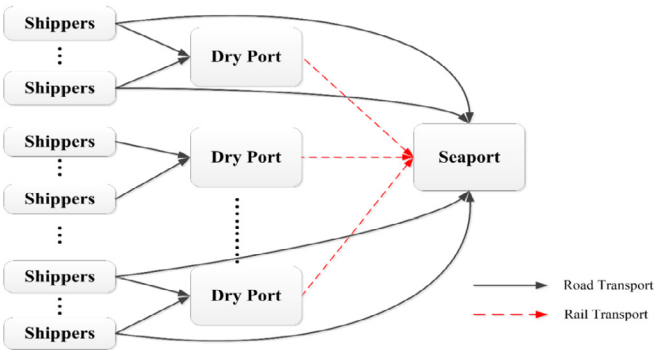


Fig. 1. A seaport-dry port network design.

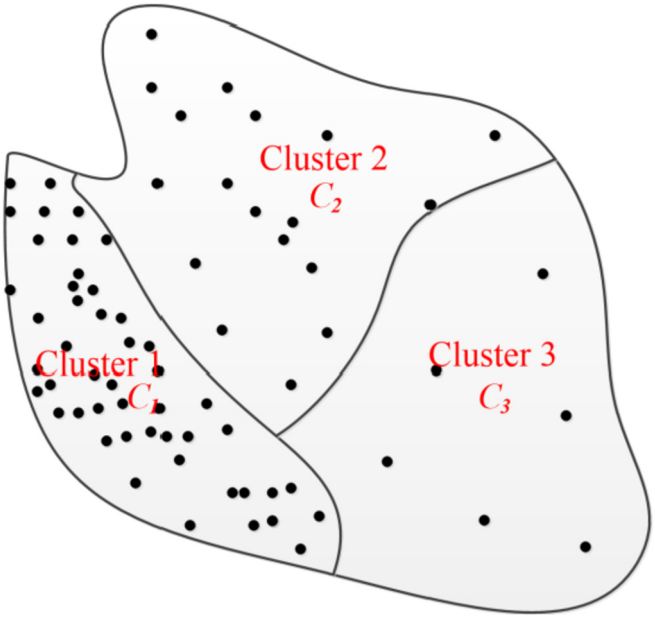


Fig. 2. Coupling.

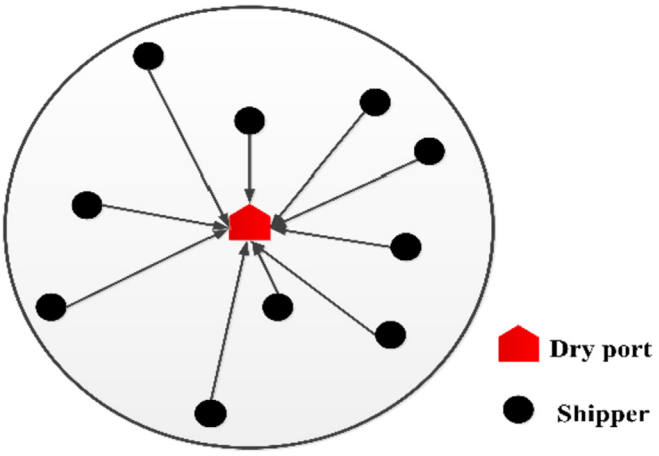


Fig. 3. Dry port service area.

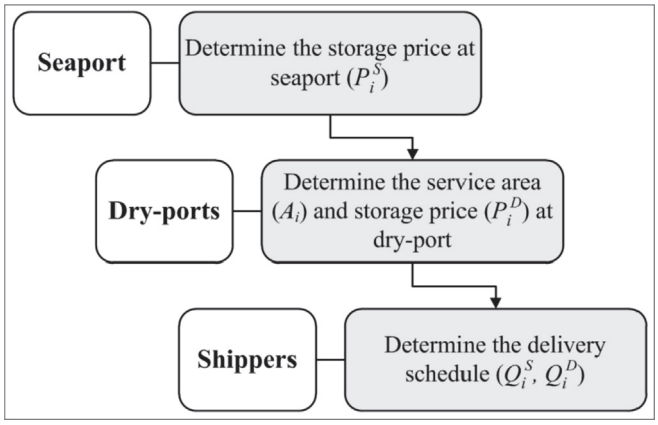


Fig. 4. Decision-making scheme.

entity makes decisions independently (Fig. 4). The seaport first determines the storage price to maximize their cost. In response, dry ports determine the service areas and storage prices to maximize the total network profit. Finally, the shippers determine the delivery schedule to minimize their total cost.

3.2. The model of seaport

3.2.1. Seaport's revenue

Shippers' demand in cluster i for transferring containers directly to seaport is$D_i^S = a_i^S - b_i^S P_i^S$, where $a_i^S > 0$ and $b_i^S > 0$. Total revenue at seaport (TR^S) includes two elements. The first element is the seaport' profit obtained through the provision of warehouse services to shippers, which is obtained by multiplying the storage price per container (P_i^S) and the total shippers demand (D_i^S) as follows$\sum_{i=1}^{N} P_i^S D_i^S$.

The second element is the dry port's profit obtained through the provision of transportation services to shippers which is calculated based on the number of delivery containers. Let $W_f$be the fixed cost per outbound container from shipper to seaport paid by shippers; $W_v$is the variable cost per outbound container from shipper to seaport paid by shippers. The seaport revenue obtained from transportation service is$\sum_{i=1}^{N}(W_f + W_v Q_i^S)\frac{D_i^S}{Q_i^S}$.

The total seaport revenue is calculated as follows

$$TR^S = \sum_{i=1}^{N} P_i^S D_i^S + \sum_{i=1}^{N}\left(W_f + W_v Q_i^S\right)\frac{D_i^S}{Q_i^S} \quad (1)$$

3.2.2. Seaport's cost components

- *The seaport operating cost:* cost for operating the seaport per unit time is $TF^S = F$.
- *The total transportation cost*

Let $M_f$be the fixed cost per outbound container from shipper to seaport paid by seaport; $M_v$is the variable cost per outbound container from shipper to seaport paid by seaport. This cost is calculated based on the number of delivery containers as follows

$$TT^S = \sum_{i=1}^{N}\left(M_f + M_v Q_i^S\right)\frac{D_i^S}{Q_i^S}, \quad (2)$$

where $M_f < W_f$ and $M_v \leq W_v$.

- *The holding cost*

Seaport is assumed to warehouse only the containers which are transferred directly from shipper (Q_i^S). The containers transferred through dry ports (Q_i^D) are lift onto board immediately when arriving. The total inventory holding cost at seaport is obtained by multiplying the holding costs and the average inventory of containers per cycle time at the seaports,$Q_i^S/2$,

$$TH^S = \sum_{i=1}^{N} h^S \frac{Q_i^S}{2}. \quad (3)$$

- *The carbon emission cost*

Trucks are used to transport the containers from shippers to the seaport. Let e_0 be the carbon emissions when the truck is empty, and$e_v$be the variable emission factor. The carbon emissions per unit time due to logistics is calculated as follows $(e_f + e_v Q_i^S)\frac{D_i^S}{Q_i^S}$(Hua et al., 2011).

The total carbon emission from warehousing Q_i^Scontainers at seaport is calculated based on the average inventory of containers per cycle time at the seaport, $\left(g_f^S + g_v \frac{Q_i^S}{2}\right)$where $g_f^S$is the fixed carbon emissions when seaport is empty, and g_v is the variable emission factor in holding containers at a seaport (Hua et al., 2011).

The total carbon cost from transportation and warehousing at seaport is

$$TT_{ce}^S = \sum_{i=1}^{N} C_{ce}\left(e_f + e_v Q_i^S\right)\frac{D_i^S}{Q_i^S} + \sum_{i=1}^{N} C_{ce}\left(g_f^S + g_v \frac{Q_i^S}{2}\right). \quad (4)$$

- *The transportation congestion cost*

Let β be the marginal external cost of congestion, which varies with the type of road as well as the volume and speed of traffic. This parameter can be determined by congestion cost estimates for

various vehicles (FHWA, 1997). It is assumed that trucks are used to ship the containers from shippers to seaport. The number of trucks is given by Q_i^S/k, where the value of k depends on the type of container. A truckload usually includes two 20-foot containers (1 TEU, $k=2$) or one 40-foot container (2 TEU, $k=1$). Transportation congestion cost is calculated as follows:

$$TT_C^S = \sum_{i=1}^{N} \beta d_i^S \frac{Q_i^S}{k}. \quad (5)$$

3.3. The model of dry ports

3.3.1. Dry ports' revenue

Shipper's demand in cluster i for transferring containers through dry port is $D_i^D = a_i^D - b_i^D P_i^D$, where $a_i^D > 0$ and $b_i^D > 0$. Total revenue at dry port (TR^D) includes two elements. The first element is the dry ports' profit obtained through the provision of warehouse services to shippers, which is calculated by multiplying the storage price per container (P_i^D) and the total shippers demand for containers transferred through dry port (D_i^D) as follows $\sum_{i=1}^{N} P_i^D D_i^D$.

The second element is the dry port's profit obtained through the provision of transportation services to shippers. The delivery fee includes the delivery charge by dry port for containers transferred from shippers to the dry ports and from the dry port to the seaport. We first obtain the total transportation fee paid by shippers based on the average distance between shippers and dry ports as follows $\sum_{i=1}^{N} W^D f_r \sqrt{A_i} D_i^D$. Where f_r is a constant depending on the distance metric and shape of the dry port service region, the average distance travelled by item is $f_r\sqrt{A_i}$ (Dasci and Verter, 2001); W^D is the delivery cost per distance per container.

Let C_f be the fixed cost per outbound container from dry port to seaport and W^S be the variable cost per item for each outbound container from dry port to seaport. Because the numbers of containers which are transferred from dry ports to seaport are large enough, the transportation fee from dry port to seaport is calculated based on the number of containers as follows $\sum_{i=1}^{N} (C_f + W^S Q_i^D) \frac{D_i^D}{Q_i^D}$.

The total dry ports' revenue is calculated as follows

$$TR^D = \sum_{i=1}^{N} P_i^D D_i^D + \sum_{i=1}^{N} W^D f_r \sqrt{A_i} D_i^D + \sum_{i=1}^{N} \left(C_f + W^S Q_i^D\right) \frac{D_i^D}{Q_i^D}. \quad (6)$$

3.3.2. Cost components

All cost functions are also modelled using a continuous approximation technique as follows.

- *Total facility cost TF* for renting and operating the dry port per unit time is formulated as follows:

$$TF^D = \sum_{i=1}^{N} f \frac{C_i}{A_i}. \quad (7)$$

where f is the facility cost per unit time and C_i/A_i is the number of dry ports in cluster i.

- *Transportation paid by dry ports*

Transportation cost from shippers to dry ports is obtained by multiplying transportation cost L^D by the average distance and the retailer demand in cluster i as follows $\sum_{i=1}^{N} (L^D f_r \sqrt{A_i} D_i^D)$.

Let L^S be the variable cost per item for each inbound container from the dry port to the seaport. Transportation cost from dry port to seaport is given by $\sum_{i=1}^{N} (L_f + L^S Q_i^D) \frac{D_i^D}{Q_i^D}$.

Total transportation cost paid by dry ports is

$$TT^D = \sum_{i=1}^{N} \left(L^D f_r \sqrt{A_i} D_i^D\right) + \sum_{i=1}^{N} \left(L_f + L^S Q_i^D\right) \frac{D_i^D}{Q_i^D}. \quad (8)$$

- *Inventory holding cost*

Dry ports work as distribution centers, depots for containers and chassis, warehouses and logistical service providers. However, they are not responsible for fulfilling when the items are out of stock. The storages are not allowed in our model. Total inventory holding cost is obtained by multiplying the number of dry ports, the holding costs and the average inventory of containers per cycle time at the dry ports, $Q_i^D/2$,

$$TH^D = \sum_{i=1}^{N} \frac{C_i}{A_i} h^D \frac{Q_i^D}{2}. \quad (9)$$

- *Total external cost of rail transportation* which includes accidents (fatalities, injures, and property damage), emissions air pollution and greenhouse gases, and noise is calculated as follows

$$TE^D = \sum_{i=1}^{N} C_a d_i^D \left(e_{tr} + e_v Q_i^D\right) \frac{D_i^D}{Q_i^D} \quad (10)$$

where $(e_{tr} + e_v Q_i^D) \frac{D_i^D}{Q_i^D}$ is the total weight Q_i^D containers associated with rail; e_{tr} is the train weight when the train is empty; e_v is variable amount of weight of loaded truck/train (per container); d_i^D is the average distance from dry port in cluster i to seaport; C_a is the rail external cost per ton per km.

- *The total carbon cost associated with containers transferred from shippers to dry ports*

Trucks are used to transport the containers from shipper to dry port. Let $(e_0 + e_v Q_i^D) \frac{D_i^D}{Q_i^D}$ be the total weight of Q_i^D containers, where e_0 is the truck weight when the truck is empty, and e_v variable amount of weight of loaded truck/train (per container). Thus, the truck external cost per weight per distance due to road transport is calculated as follows:

$$TE^D = \sum_{i=1}^{N} C_u f_r \sqrt{A_i} \left(e_0 + e_v Q_i^D\right) \frac{D_i^D}{Q_i^D}. \quad (11)$$

- *The total carbon cost associated with warehousing at dry ports*

The carbon emissions from warehouse are assumed to be linear in the inventory. As the previous, the average inventory of containers per cycle time at the dry port is $Q_i^D/2$. So that the total

carbon emissions from warehousing at dry port per cycle time is$g_f + g_v\frac{Q_i^D}{2}$, where $g_f$is the fixed carbon emissions, and g_v is the variable emission factor in holding containers at a dry port (Hua et al., 2011). Thus, the total carbon cost associated with warehousing at dry ports is

$$TT_{ce}^{D} = \sum_{i=1}^{N} C_{ce}\left(g_f + g_v\frac{Q_i^D}{2}\right). \tag{12}$$

3.4. The model of shipper

Shippers determine the shipping schedule with aiming at total cost minimization. Note that the total number of container deliveries in each delivery from shippers to dry port and seaport is given,$Q_i = Q_i^S + Q_i^D$. Shippers' cost components are formulated as follows.

The cost of handling containers is known as storage charge. The total storage charge is formulated as follows:

$$TS = \sum_{i=1}^{N} P_i^D D_i^D + \sum_{i=1}^{N} P_i^S D_i^S. \tag{13}$$

As detailed in the previous section, the total transportation cost is

$$TT^{Sh} = \sum_{i=1}^{N}\left(\mathrm{W}^D f_r\sqrt{A_i}D_i^D\right) + \sum_{i=1}^{N}\left(C_f + \mathrm{W}^S Q_i^D\right)\frac{D_i^D}{Q_i^D} + \sum_{i=1}^{N}\left(W_f + W_v Q_i^S\right)\frac{D_i^S}{Q_i^S}. \tag{14}$$

4. Solution Approach-A game-theoretical decision-making scheme

This section illustrates the solution procedure used in the proposed model. Seaport and dry ports make their decisions for optimizing based on the reactions obtained from shippers.

4.1. Shipper reaction

From Eqs. 13 and 14, the shippers' objective function is formulated as follows.

$$\begin{aligned}\text{Min } TC &= TS + TT^{sh}\\ &= P_i^S D_i^S + P_i^D D_i^D + \sum_{i=1}^{N}\left(\mathrm{W}^D f_r\sqrt{A_i}D_i^D\right) + \sum_{i=1}^{N}\left(C_f + \mathrm{W}^S Q_i^D\right)\frac{D_i^D}{Q_i^D} + \sum_{i=1}^{N}\left(W_f + W_v Q_i^S\right)\frac{D_i^S}{Q_i^S}\end{aligned} \tag{15}$$

Subject to. $Q_i = Q_i^S + Q_i^D$

The problem analysed here is to determine the optimal delivery schedule (Q_i^S, Q_i^D) to minimize total shippers' cost, $TC(Q_i^S, Q_i^D)$. Given the value of Q_i, the number of container deliveries from shippers to seaport is$Q_i^S = Q_i - Q_i^D$. Substituting$Q_i^S = Q_i - Q_i^D$into Eq. (15) yields

$$\begin{aligned}TC = {} & P_i^S D_i^S + P_i^D D_i^D + \sum_{i=1}^{N}\left(\mathrm{W}^D f_r\sqrt{A_i}D_i^D\right) + \sum_{i=1}^{N}\left(C_f + \mathrm{W}^S Q_i^D\right)\frac{D_i^D}{Q_i^D}\\ & + \sum_{i=1}^{N}\left(W_f + W_v\left(Q_i - Q_i^D\right)\right)\frac{D_i^S}{\left(Q_i - Q_i^D\right)}\end{aligned} \tag{16}$$

The function of TCbecomes a non-linear function of 1N variables (Q_i^D, for $i = 1, 2, \ldots, N$). To solve the problem, we first take the second-order derivatives of Eq. (17) with respective toQ_i^D yields

$$\frac{\partial^2 TC\left(Q_i^D\right)}{\partial\left(Q_i^D\right)^2} = \frac{2\left(\left(a_i^D - b_i^D P_i^D\right)\left(Q_i^D - Q_i\right)^3 C_f - \left(a_i^S - b_i^S P_i^S\right)\left(Q_i^D\right)^3 W_f\right)}{\left(Q_i^D\right)^3\left(Q_i^D - Q_i\right)^3} > 0 \tag{17}$$

TC is a convex function with respect to Q_i^D . This means that the optimal Q_i^{D*}can be obtained by solving $\frac{\partial TC(Q_i^D)}{\partial Q_i^D} = 0$:

$$Q_i^{D*} = \frac{\left(a_i^D - b_i^D P_i^D\right)Q_i C_f + \sqrt{\left(a_i^D - b_i^D P_i^D\right)\left(a_i^S - b_i^S P_i^S\right)Q_i C_f W_f}}{\left(a_i^D - b_i^D P_i^D\right)C_f - \left(a_i^S - b_i^S P_i^S\right)W_f} \tag{18}$$

The optimal Q_i^{S*}can be obtained by$Q_i^{S*} = Q_i - Q_i^{D*}$:

$$Q_i^{S*} = Q_i - \frac{\left(a_i^D - b_i^D P_i^D\right)Q_i C_f + \sqrt{\left(a_i^D - b_i^D P_i^D\right)\left(a_i^S - b_i^S P_i^S\right)Q_i^2 C_f W_f}}{\left(a_i^D - b_i^D P_i^D\right)C_f - \left(a_i^S - b_i^S P_i^S\right)W_f} \tag{19}$$

4.2. Dry-port decision-making

The variables in this model include the service areasA_i, and storage prices, $P_i^D$for dry ports in cluster i. From Eqs. 6–12, the dry ports' objective function is formulated as follows.

$$\begin{aligned}\text{Min } PR^D &= TR - TF - TT^D - TH - TE^D - TT^D_{ce} = \sum_{i=1}^{N} P_i^D D_i^D + \sum_{i=1}^{N} W^D f_r \sqrt{A_i} D_i^D + \sum_{i=1}^{N} \left(C_f + W^S Q_i^D\right) \frac{D_i^D}{Q_i^D} - \sum_{i=1}^{N} f \frac{C_i}{A_i} - \sum_{i=1}^{N} \left(L^D f_r \sqrt{A_i} D_i^D\right) \\ &- \sum_{i=1}^{N} \left(L_f + L^S Q_i^D\right) \frac{D_i^D}{Q_i^D} - \sum_{i=1}^{N} \frac{C_i}{A_i} h^D \frac{Q_i^D}{2} - \sum_{i=1}^{N} C_a d_i^D \left(e_{tr} + e_v Q_i^D\right) \frac{D_i^D}{Q_i^D} - \sum_{i=1}^{N} C_u f_r \sqrt{A_i} \left(e_0 + e_v Q_i^D\right) \frac{D_i^D}{Q_i^D} - \sum_{i=1}^{N} C_{ce}^D \left(g_f + g_v \frac{Q_i^D}{2}\right)\end{aligned} \tag{20}$$

subject to $W^D > L^D$; $W^S > L^S$; $D_i^D = a_i^D - b_i^D P_i^D$; $h < P_i^D$; $Q_i^S + Q_i^D = Q_i$.

PR^D is a concave function with respect to A_i and P_i^D (Appendix A) that means a strict global maximize of PR^D could be found and called as A_i^* and P_i^{D*}. To determine the optimal value for the decision variables, we take the first-order derivative of PR^D with respect to A_i and P_i^D and set the results at zero ($\frac{\partial PR^D}{\partial A_i} = 0$, $\frac{\partial PR^D}{\partial P_i^D} = 0$), thereby obtaining the following:

$$A_i^* = \frac{\left(\left(2F + h^D Q_i^D\right)^2\right)^{1/3} C_i^{2/3}}{\left(\left(a_i^D - b_i^D P_i^D\right)^2 \left(L^D - W^D\right)^2\right)^{1/3} f_r^{2/3}} \tag{21}$$

$$P_i^{D*} = \frac{a_i^D}{2b_i^D} + \frac{C_a d_i^D e_v + L^S - W^S + C_u e_v f_r + L^D f_r \sqrt{A_i} - W^D f_r \sqrt{A_i}}{2} + \frac{C_u e_0 f_r + L_f - C_f + C_a d_i^D e_{tr}}{2Q_i^D} \tag{22}$$

4.3. Seaport decision-making

From Eqs. (1)–(5), the seaport's objective function is formulated as follows.

$$\begin{aligned}\text{Max } PR^S &= TR^S - TF^S - TT^S - TH^S - TT^S_{ce} - TT^S_C \\ &= \sum_{i=1}^{N} P_i^S D_i^S + \sum_{i=1}^{N} \left(W_f + W_v Q_i^S\right) \frac{D_i^S}{Q_i^S} - F - \sum_{i=1}^{N} \left(M_f + M_v Q_i^S\right) \frac{D_i^S}{Q_i^S} - \sum_{i=1}^{N} h^S \frac{Q_i^S}{2} - \sum_{i=1}^{N} C_{ce} \left(e_f + e_v Q_i^S\right) \frac{D_i^S}{Q_i^S} - \sum_{i=1}^{N} C_{ce} \left(g_f^S + g_v \frac{Q_i^S}{2}\right) - \sum_{i=1}^{N} \beta d_i \frac{Q_i^S}{k}\end{aligned} \tag{23}$$

The function of PR^S is a non-linear function of 1N variables (P_i^S, for $i = 1, 2, \ldots, N$). To solve the problem, we first take the second-order derivatives of Eq. (23) with respective to P_i^S yields

$$\frac{\partial^2 PR^S\left(P_i^S\right)}{\partial\left(P_i^S\right)^2} = -2b_i^S < 0 \tag{24}$$

$PR^S(P_i^S)$ is a concave function of P_i^S . This means that the optimal P_i^{S*} can be obtained by solving $\frac{\partial PR^S(P_i^S)}{\partial P_i^S} = 0$:

$$P_i^{S*} = \frac{C_{ce} e_f + M_f - W_f}{2Q_i^S} + \frac{a_i^S + b_i^S C_{ce} e_v + b_i^S M_v - b_i^S W_v}{2b_i^D} \tag{25}$$

4.4. Algorithm procedure

As a summary, the decision-making scheme is as follows: Seaport first determines the storage price (P_i^S) to maximize its profit; Secondly, the dry ports determine the service areas (A_i) and storage prices (P_i^D) to maximize their total network profit; finally, the shippers consequently determine the delivery schedule (Q_i^S, Q_i^D) to minimize their total cost. The above solution procedure is summarized as follows.

Step 1: Calculate the optimal $P_i^{S*}(Q_i^{S*})$ with Eq. (25)

Step 2: Substitute $P_i^{S*}(Q_i^{S*})$ into Eq. (18) to obtain the optimal. $Q_i^{D*}(P_i^{D*})$

Step 3: Substitute $Q_i^{D*}(P_i^{D*})$ into Eq. (21) to obtain the optimal. $A_i^*(P_i^{D*})$

Step 4: Substitute $Q_i^{D*}(P_i^{D*})$ and $A_i^*(P_i^{D*})$ into Eq. (22) to obtain the optimal. P_i^{D*}

Step 5: Given value of P_i^{D*}, calculating the optimal value of A_i^*, Q_i^{D*}, Q_i^{S*} and. P_i^{S*}

Step 6: Calculate the optimal value of $PR^S(P_i^{S*})$, $PR^D(A_i, P_i^{S*})$, and $TC(Q_i^{D*}, Q_i^{S*})$.

5. Numerical analysis

5.1. Example

In this section, a three-echelon supply chain network consisting of a seaport, dry ports, and shippers is considered. The service area of the seaport covers three clusters: they are Cluster 1 ($C_1 = 100000\,\text{km}^2$), Cluster 2 ($C_2 = 80000\,\text{km}^2$) and Cluster 3 ($C_3 = 60000\,\text{km}^2$), respectively, in our examples. Assume that the coefficient of demand in three clusters are similar in both seaport

Table 1
The optimal decisions and performance of seaport, dry ports, and shippers.

	Cluster i	1	2	3
Seaport decision	$P_i^{S*}(USD)$	185.365	256.077	195.633
	$PR^S(USD)$	179,922.872		
Dry Ports decision	$P_i^{D*}(USD)$	135.162	146.678	143.861
	A_i (km^2)	38,369.2	32,062.4	21,611.2
	Number of dry ports	3	2	3
	$PR^D(USD)$	848,914.168		
Shippers decision	Q_i^S(containers)	121.876	26.7	118.21
	Q_i^D(containers)	178.124	153.3	131.79
	$TC(USD)$	2,057,935.179		

and dry port, the value of parameters are $a_1^D = a_1^S = 1800; a_2^D = a_2^S = 1600 a_3^D = a_3^S = 1900; b_1^D = b_1^S = 5 b_2^D = b_2^S = 4; b_3^D = b_3^S = 5$. The facility cost for opening and operating each seaport and dry port are $F = 100000$ (USD) and $f = 5000$ (USD). The cost of transportation and holding containers are $C_f = 1000$ (USD/shipment); $C_u = 0.0069$ (USD/ton/km); $C_a = 0.0017$ (USD/ton/km); $C_{ce} = 5.4$ (USD/m3); $h^S = 120$ (USD/TEU/month); $h^D = 100$ (USD/TEU/month); $L^S = 50$ (USD/TEU); $L^D = 20$ (USD/TEU); $L_f = 800$ (USD/shipment); $M_f = 1200$ (USD/shipment); $M_v = 120$ (USD/TEU); $W^S = 70$ (USD/TEU); $W^D = 36$ (USD/TEU); $W_f = 1800$ (USD/shipment); $W_v = 180$ (USD/TEU/shipment). The amount carbon emission warehousing in dry port and seaport are $g_f^S = 10000$ (m^3/seaport/month) and $g_f^D = 1000$ (m^3/seaport/month). The weight of truck, train, and container are $e_f = 500$ (t/truck); $e_{tr} = 5400$ (t/train); $e_v = 2.5$ (t/TEU). The distance from shippers and dry port to seaport in each cluster i are $d_1^D = 620$ (km), $d_2^D = 399$ (km), $d_3^D = 201$ (km), $d_1^S = 800$ (km), $d_2^S = 615$ (km), and $d_3^S = 335$ (km).

Mathematica 7.0 is used to run our algorithm. Table 1 presents the optimal decisions and performance of seaport, dry ports, and shippers wherein profits are maximized and costs minimized.

In our example, three kind of dry ports (distant, midrange and close dry ports) are considered. The Cluster one is located farthest from the seaport and Cluster three is most closed to the seaport. Fig. 5 shows that shippers tend to choose direct route to seaport in Cluster three. Although the storage price at seaport is higher, the cost saving from transportation service encourages the shippers to transfer containers directly to seaport.

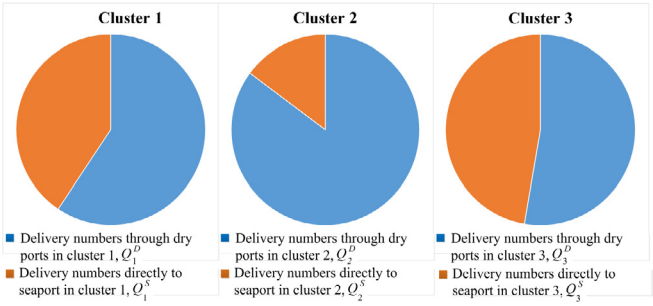


Fig. 5. The different delivery schedule of three clusters.

5.2. 5.2Sensitivity analysis of major parameters

5.2.1. Varying storage holding cost at seaport and dry port (h^S, h^D)

Firstly, the holding cost at seaport and dry port (h^S, h^D) are varied, but keep other parameters the same as the base scenario. Table 2 shows the impact of the holding cost at seaport and dry port on shippers' cost, seaport and, dry port profits. Shippers have to pay for higher cost while the profits of dry port and seaport decrease. This finding provides a guideline for the dry port operator that reducing the holding cost would make a higher profit for dry ports and seaport; also decrease the shippers' cost.

Fig. 6 shows the impact of the holding cost at the seaport and dry port on the decision variables. Seaport first tends to increase its storage price (P_i^S) to compensate for an increase the storage holding cost (h^S). As the followers, dry ports set a lower storage price (P_i^D) to get more order quantities from shippers to compensate for increasing the storage holding cost (h^D). Shippers choose the indirect route through dry ports instead of the direct route to the seaport. Thus, the number of containers deliveries through dry ports increases and the number of containers deliveries directly to dry ports decreases.

5.2.2. Varying carbon cost (C_{ce})

In this sub-section, we vary the carbon cost (C_{ce}), but keep other parameters the same as the base scenario. Table 3 shows the impact of the carbon cost on shippers' cost, seaport, and dry port profits. Shippers have to pay for higher cost while the profits of dry port

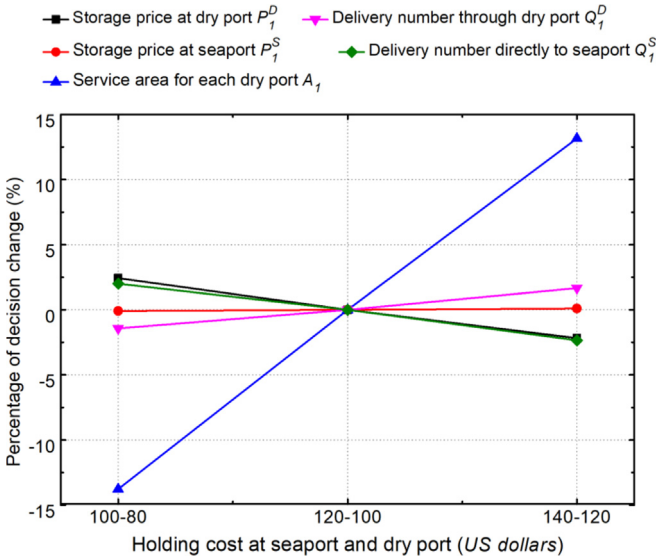


Fig. 6. The influence of the holding cost at seaport and dry ports (in cluster 1).

Table 2
Optimal cost and profits for shippers, dry ports, and seaport with varying storage holding cost (in cluster 1).

Varying storage holding cost (USD)			
(h^S, h^D)	$TC_1(Q_1^S, Q_1^D)$	$PR_1^D(A_1, P_1^D)$	$PR_1^S(P_1^S)$
(100, 80)	674,540.765	261,662.662	61,914.399
(120,100)	683,670.591	260,230.869	63,484.620
(140, 120)	691,797.444	258,434.009	64,696.030

Table 3
Optimal cost and profits for shippers, dry ports, and seaport with varying carbon cost (in cluster 1).

Varying carbon cost (USD)			
C_{ce}	$TC_1(Q_1^S, Q_1^D)$	$PR_1^D(A_1, P_1^D)$	$PR_1^S(P_1^S)$
4.5	691445.6	259441.3	76925.04
5.4	691797.4	258434.0	64696.03
7	692210.2	256602.8	42946.46

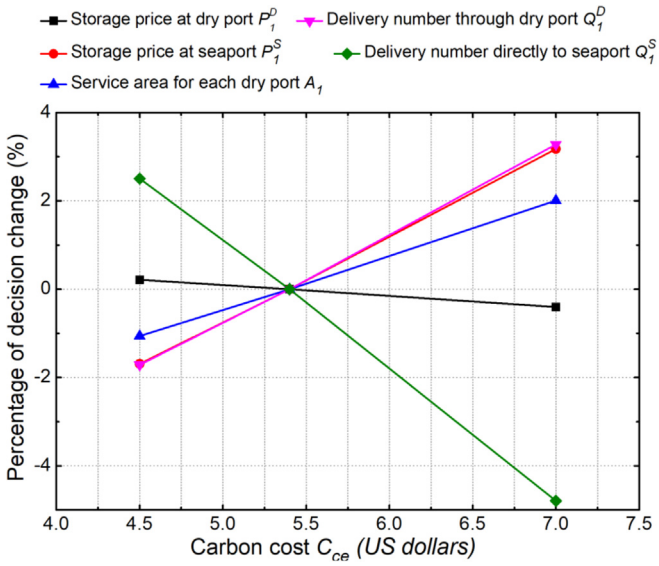


Fig. 7. The influence of the carbon cost (in cluster 1).

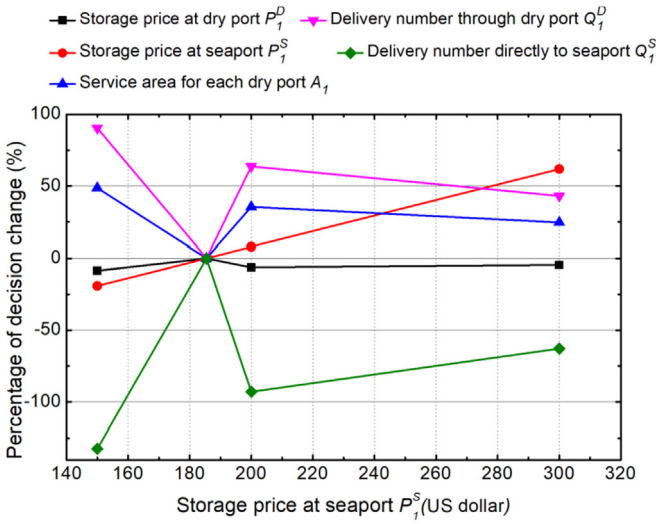


Fig. 8. The influence of the storage cost at seaport (in cluster 1).

and seaport decrease. Seaport' profit decrease sharply while the carbon cost increases. This finding illustrates that the opening of dry port really helps seaport to reduce the environmental effect.

When carbon emissions cost (C_{ce}) increases, seaport first tends to charge more the storage price (P_i^S) to compensate for an increase in carbon cost (Fig. 7). Later, dry ports have to less opening dry port to reduce the effect of increasing the carbon cost associated with warehousing at dry port, thus, the service area of each dry port (A_i) increases. However, they also decrease their storage price (P_i^D) to get more order quantities from shippers and keep the total profit in good proportion. Finally, shippers make their decisions based on storage price at the seaport and dry port. In this case, transferring containers through dry ports is a better option for shippers.

5.2.3. *Varying storage price at seaport* (P_i^S)

In our model, seaport plays as a leader, its decision-making affects to followers' decision. In this sub-section, the storage price at seaport (P_i^S) is varied, but keep other parameters the same as the base scenario. Table 4 shows the impact of P_i^S on shippers' cost, seaport, and dry port profits. Shippers' cost tends to fluctuate when the storage price at seaport increases. Shippers may adjust their shipping schedule to fit with the changing of store price at the seaport and dry ports. In a different way, the profit of dry ports increases slightly. When the storage price at seaport increases the dry ports can earn more profits by getting more order quantities from shippers (Fig. 8).

Fig. 8 illustrates the impact of varying the storage price at seaport on other decision variables. As the storage price at seaport increases, the storage price at dry port first decreases to get more orders from shippers; then increases to get more revenue. The changing of the storage price at dry port is not too much to guarantee that their profit increases. In response to leader' decision, shippers change their shipping schedule to minimize their total cost. Fig. 8 also shows that the number of containers deliveries through dry port or directly to seaport varies on the changing of the storage price.

Table 4
Optimal cost and profits for shippers, dry ports, and seaport with varying P_i^S (in cluster 1).

Varying storage price at seaport (USD)			
P_i^S	$TC_1(Q_1^S, Q_1^D)$	$PR_1^D(A_1, P_1^D)$	$PR_1^S(P_1^S)$
150.00	710477.99	256919.25	190683.91
185.40	691797.44	258434.01	64696.03
200.00	706380.63	258563.89	−250724.35
300.00	631469.22	259409.05	81592.15

6. Conclusions

In this study, a CA approach is used to design a seaport-dry port network under carbon emission consideration. The results show that the proposed method could address the seaport-dry port network problems well. A game-theoretical decision-making scheme is applied to help the seaport determine the storage price; the government or companies determine where to set up the dry ports and how many dry ports should be opened in each area; the shippers make decisions on delivery schedule. The closed form of all decision variables are found directly by non-linear optimization technique.

Based on the numerical analysis, this study provides some interesting and useful managerial insights as follows. First, the shippers who are next to seaport prefer the direct route to seaport than the route through dry port. In contrast, the shippers who locate far from seaport tend to choose the indirect route through the dry ports. For example, in our example, the large service areas (cluster 1 and cluster 2) yields the higher rate of containers transferred through the dry ports ($Q_1^D/Q_1^S = 1.46 \text{and} Q_2^D/Q_2^S = 5.74$) while this rate in cluster 3 is $Q_3^D/Q_3^S = 1.11$. With the big countries such as America, India, China, the opening of dry ports gets more benefits than others. However, with small countries but have developed rail systems like Euro countries, the development of dry port system can help them to reduce the carbon and congestion cost from road transport. For environmental considerations, the opening and operating of dry ports is a good choice for all developed and developing countries.

The numerical results also show that the profits of the seaport and dry ports decrease while the costs assumed by shippers increase as the holding costs increase. Seaport or dry ports should apply new technology to decrease the transportation cost by train, use green power to operate their operations and transportation or decrease their holding cost to get more customers. The results also show that the carbon cost plays an important role in the decision making process for the three players. When the carbon cost per unit increases, shippers tend to choose the indirect route through dry port than the direct route to the seaport. The number of containers transferred directly to seaport decreases by 7 percentages when the carbon cost increases from 4.5USD to 7USD per unit. In addition, when the carbon cost increases, the total profit of seaport decreases sharply (about 44 percentage) while the total profits of dry ports make a slight drop (about 1.1 percentage). The reason is to decrease the carbon cost of road transport. In practice, railroad transports are four times more fuel efficient than road transportation (AoAR, 2017). Therefore, the development of dry port concept with multimodal transport is one of the important ways to reduce the carbon emission from road transport and protect the environment.

Future research could focus on developing a seaport-dry-port network design under uncertain demand, and apply the Internet of Things (IoT) to its operation. In addition, the competition between seaports is also an interesting topic that could be considered in the future works.

Appendix A

From total dry port's profit functionPR^D, the Hessian matrix of PR^Dcan be formulated as

$$H = \begin{vmatrix} \frac{\partial^2 PR^D}{\partial (A_i)^2} & \frac{\partial^2 PR^D}{\partial A_i \partial P_i^D} \\ \frac{\partial^2 PR^D}{\partial P_i^D \partial A_i} & \frac{\partial^2 PR^D}{\partial \left(P_i^D\right)^2} \end{vmatrix} = \begin{vmatrix} \frac{-4\left(2f + h^D Q_i^D\right) C_i + \left(a_i^D - b_i^D P_i^D\right)\left(L^D - W^D\right) A_i^{3/2} f_r}{4A_i^3} & \frac{b_i^D \left(L^D - W^D\right) f_r}{2\sqrt{A_i}} \\ \frac{b_i^D \left(L^D - W^D\right) f_r}{2\sqrt{A_i}} & -2b_i^D \end{vmatrix} \tag{A1}$$

To verify whether the matrix given in Eq. (A1) is positive or not, we evaluate its principal minors:

$$D_1 = \frac{\partial^2 PR^D}{\partial (A_i)^2} = \frac{-4\left(2f + h^D Q_i^D\right) C_i + \left(a_i^D - b_i^D P_i^D\right)(L^D - W^D) A_i^{3/2} f_r}{4A_i^3} < 0 \forall A_i \tag{A2}$$

$$D_2 = \frac{b_i^D \left(8\left(2f + h^D Q_i^D\right) C_i - \left(L^D - W^D\right) A_i^{3/2} f_r \left(2\left(a_i^D - b_i^D P_i^D\right) - b_i^D \left(L^D - W^D\right)\sqrt{A_i} f_r\right)\right)}{4A_i^3} \tag{A3}$$

$$D_2 > 0 \forall A_i, P_i^D$$

Thus, we know that the Hessian Matrix is negative definite since all principle minors are negative equations (A1), (A2), and (A3) are positive. The total cost for shippers PR^D is concave function.

Appendix B. –Nomenclature

Indexes	
i	Set of power plant indexes, $i = 1, 2\ldots, N$
Parameters	
$a_i^D, b_i^D, a_i^S, b_i^S$	The coefficient of demand function
C_i	Service area for cluster i
C_f	Fixed cost per outbound container from dry port to seaport paid by shipper
C_u	Truck external cost per unit
C_a	The rail external cost per ton per km
C_{ce}	Carbon emission cost per unit paid by dry port
D_i^D	Shipper's demand for containers deliveries through dry port in cluster i
D_i^D	Shipper's demand for containers deliveries directly to seaport in cluster i
d_i^S	The average distance from shippers in cluster i to seaport
d_i^D	The average distance from dry port in cluster i to seaport
e_f	Weight of empty truck (weight unit per truck)
e_{tr}	Weight of empty train (weight unit per train)
e_v	Variable amount of weight of loaded truck/train (per container)
F	Facility cost of operating per unit time for seaport
f	Facility cost of opening and operating per unit time for each dry port
f_r	A constant depending on the distance metric and shape of dry port service region
g_f^S	Fixed amount of carbon emissions of empty seaport
g_f^D	Fixed amount of carbon emissions of empty dry port
g_v	Variable amount of carbon emissions associated with warehousing per container
h^S	Inventory holding costs at seaport
h^D	Inventory holding costs at dry port
k	The number of containers per truck
L^D	Delivery fee per distance per container from shippers to the dry port (paid by dry port)
L^S	delivery fee per container from the dry port to the seaport (variable cost, paid by dry port)
L_f	The fixed cost per outbound container from dry port to seaport paid by dry port
M_f	The fixed cost per outbound container from shipper to seaport paid by seaport
M_v	The variable cost per containers per outbound from shipper to seaport paid by seaport
Q_i	The total number of containers deliveries from shippers in cluster i (in each delivery)
S	The service area of seaport
W^D	Delivery fee charged by dry port per distance per container from shippers to the dry port
W^S	Delivery fee charged by dry port per container from the dry port to seaport (variable cost)
W_f	The fixed cost per outbound container from shipper to seaport paid by shippers
W_v	The variable cost per containers per outbound from shipper to seaport paid by shippers
β	The marginal external cost of congestion
Decision Variables for seaport	
P_i^D	Storage price per container at seaport paid by shipper in cluster i
Decision Variables for dry port	
A_i	Service area of each dry port in cluster i
P_i^S	Storage price per container at dry port in cluster i
Decision Variables for dry port	
Q_i^D	Number containers deliveries from shipper in cluster i through dry port to seaport (in each delivery)
Q_i^S	Number containers deliveries directly from shipper in cluster i to seaport (in each delivery)

References

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